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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 6	
TITLE : Inner and Anonymous Classes	

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Programme

Exercise

1.Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action

```
import java.util.Scanner;
```

```
class Action {  
    void run(double distance, double fuel) {  
    }  
}
```

```
public class Vehicle {
```

```
    String make = "Toyota";
```

```
    String model = "Corolla";
```

```
    class Details {
```

```
void show() {  
    System.out.println("Vehicle: " + make + " " + model);  
}  
}
```

```
public static void main(String[] args) {  
  
    Scanner sc = new Scanner(System.in);  
  
    Vehicle car = new Vehicle();  
    Vehicle.Details printer = car.new Details();  
    printer.show();  
    System.out.print("Enter distance traveled: ");  
    double distance = sc.nextDouble();  
  
    System.out.print("Enter fuel used: ");  
    double fuel = sc.nextDouble();  
    Action mileageCalculator = new Action() {  
        void run(double d, double f) {  
            System.out.println("Mileage: " + (d / f));  
        }  
    };  
    mileageCalculator.run(distance, fuel);  
    sc.close();  
}
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```
class FoodDelivery {

    String customer = "Rahul";

    class OrderDetails {

        int orderId = 101;

        String food = "Pizza";

        int price = 300;

        void showOrder() {

            System.out.println("Customer : " + customer);

            System.out.println("Order ID : " + orderId);

            System.out.println("Food    : " + food);

            System.out.println("Price   : " + price);

        }

    }

    void showStatus() {

        Status s1 = new Status() {

            void display() {

                System.out.println("Order Confirmed");

            }

        };

        Status s2 = new Status() {

            void display() {

                System.out.println("Out for Delivery");

            }

        };

    }

}
```

```

};

Status s3 = new Status() {

    void display() {

        System.out.println("Order Delivered");

    }

};

s1.display();

s2.display();

s3.display();

}

}

class Status {

    void display() {

        System.out.println("Status");

    }

}

public class Main {

    public static void main(String[] args) {

        FoodDelivery fd = new FoodDelivery();

        FoodDelivery.OrderDetails od = fd.new OrderDetails();

        od.showOrder();

        System.out.println("\nDelivery Status:");

        fd.showStatus();

    }

}

```

Output

1.

```
Vehicle: Toyota Corolla
Enter distance traveled: 300
Enter fuel used: 15
Mileage: 20.0
```

2.

```
Customer : Rahul
Order ID : 101
Food      : Pizza
Price     : 300

Delivery Status:
Order Confirmed
Out for Delivery
Order Delivered
```

Conclusion

Thus, we successfully implemented inner classes and anonymous classes for improving encapsulation and event handling in Java.